Methods Practice SAC1 - Part B

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Notes + CAS | 10m Reading | 1h 40m Writing | 51 marks

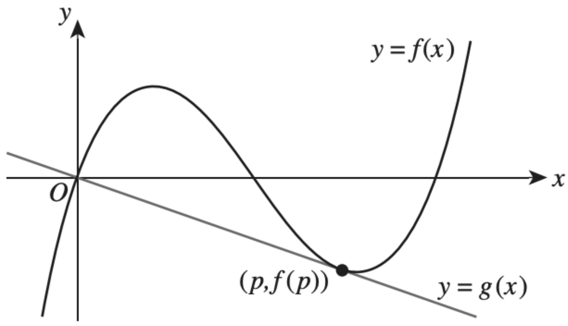
Question 1 (7 marks)

Neap 2020 Exam 2 Q1

Consider the following functions, where a is a positive constant:

$$f : R \to R, f(x) = x^3 - 6x^2 + 8x$$
$$g : R \to R, g(x) = -ax$$

Part of the graphs of f and g are shown in the diagram below.



The line $y = g(x)$ is tangent to the graph of f at $x = p$ where $p > 0$.

a. Find the coordinates of the x-intercepts of the graph of $y = f(x)$. 1 mark

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b. Show that $a = 1$ and $p = 3$. 3 marks

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c. Find the maximum vertical distance between the graphs of f and g over the interval $x \in [0, 3]$. 3 marks

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Question 2 (12 marks)

Consider the function, f_1, defined as:

$$f_1\colon \left[0, \frac{4\pi}{3}\right] \to R,\ f_1(x) = \sqrt{2}\,sin\left(x - \frac{\pi}{3}\right) + 1$$

a. Find the domain and rule of the derivative, f_1'. (2 marks)

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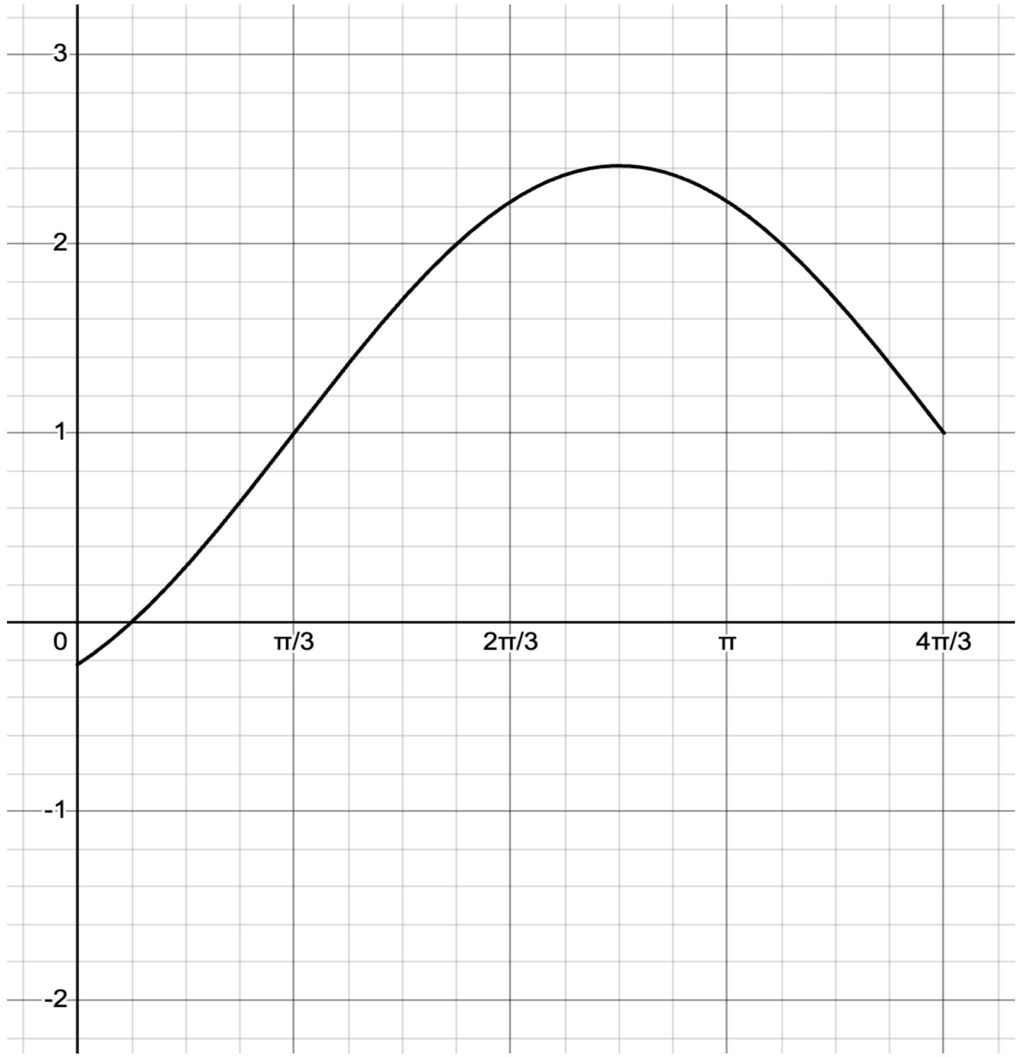
b. Sketch the graph of $y = f_1'(x)$ on the axis below. Make sure to include, if applicable, axial intercepts, endpoints and the coordinates of intersections with f_1. (4 marks)



Consider a new function, g, defined as:

$$g: [0, \frac{4\pi}{3}] \to R,\ g(x) = f_1'(x) + k$$

c. Find the values of k for which the graphs of f and g have no intersection. (2 marks)

d. Now consider f_1 without its restricted domain as the function f, where:

$$f: R \to R,\ f(x) = \sqrt{2}\, sin(x - \frac{\pi}{3}) + 1$$

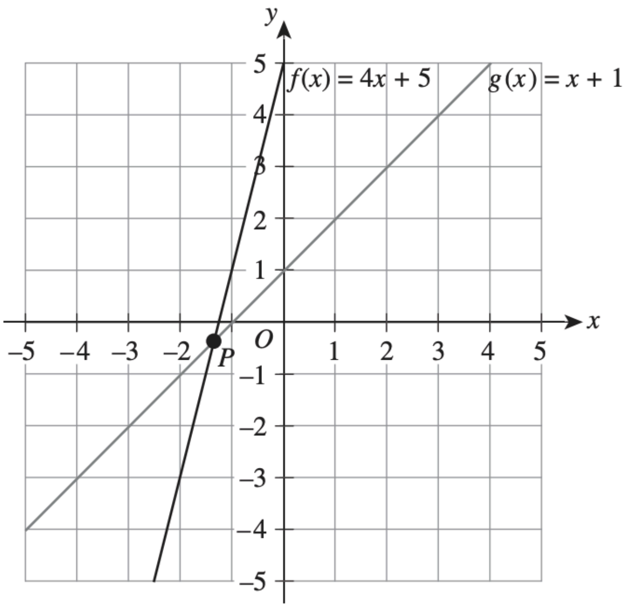
i. State the derivative, f', in the form $f'(x) = a\, sin(x + b)$ where $a, b \in R$. (1 mark)

ii. Hence, or otherwise, state the sequence of transformations required to take the graph of f to the graph of f'. (3 marks)

Question 3 (13 marks)

Neap 2020 Exam 2 Q2 (truncated)

Part of the graphs of the functions f and g are shown below where $f : R \to R, f(x) = 4x + 5$ and $g : R \to R, g(x) = x + 1$.



a. **i.** Find the coordinates of the point of intersection, P. 1 mark

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ii. Find the distance of P from the origin. 1 mark

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iii. Find the size of the acute angle between the graphs of f and g at P. Give your answer to the nearest degree. 1 mark

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A transformation $T_1 : R^2 \to R^2$ maps g to a new function h, where T_1 is given by the following:

$$(x, y) \to (ax, \tfrac{1}{a}y) \text{ where } a \in R^+$$

b. **i.** Describe the transformations that map the graph of $y = g(x)$ to $y = h(x)$. 1 mark

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ii. Hence, or otherwise, find the rule of the function $h(x)$ in terms of a. 2 marks

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c. State the value(s) of a for which there is a unique solution to the equation $f(x) = h(x)$. 2 marks

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d. Find the coordinates of the point of intersection of the graphs of $y = f(x)$ and $y = h(x)$ in terms of a. 2 marks

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e. Find the value of a for which the distance between the point of intersection of the graphs of $y = f(x)$ and $y = h(x)$ and the origin is a minimum. 3 marks

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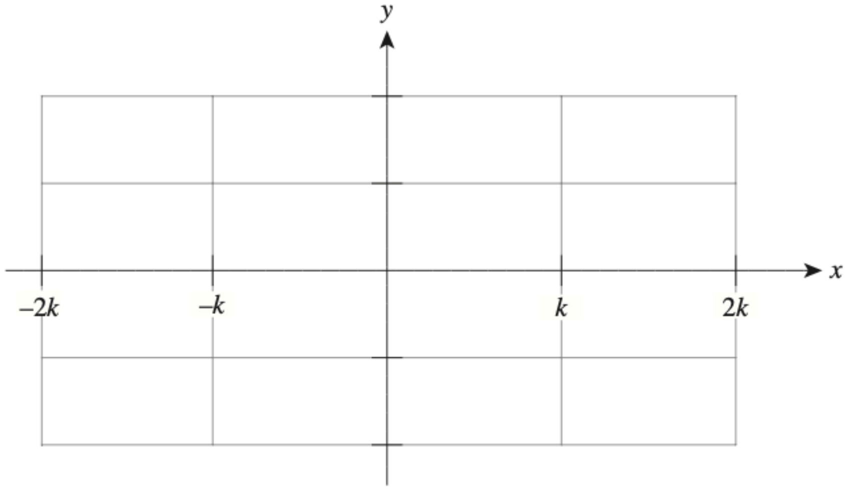
Question 4 (6 marks)

Neap 2021 Exam 2 Q2

Consider the function with the rule $f(x) = (x+k)(x-k)^2$, where $k \in R^+$.

a. Graph $y = f(x)$ on the set of axes below, showing the coordinates of all intercepts and turning points in terms of k where appropriate. 3 marks



b. Consider the equation $x^2 - k^2 = \dfrac{1}{x-k}$.

i. Show that this equation can be expressed as $(x+k)(x-k)^2 = 1$. 1 mark

ii. For what value(s) of k does the equation $x^2 - k^2 = \dfrac{1}{x-k}$ have exactly two solutions? 2 marks

Question 5 (13 marks)

Consider the function, f, defined as:

$$f\colon R \to R,\ f(x) = x^3 - 3x$$

And the formula for the 'next x-intercept' for Newton's method:

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

a. Consider the the initial guess, $x_0 = -\frac{9}{10}$.

i. Fill in the table, correct to 4 decimal places, for the values of Newton's method for the initial guess, $x_0 = -\frac{9}{10}$. (2 marks)

x_0	
x_1	
x_2	
x_3	
x_4	

ii. Which x-intercept of f is Newton's method converging to? (1 mark)

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b. Find the values of x_n for which Newton's method <u>fails</u> due to there being no next x-intercept, x_{n+1}. (2 marks)

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c. Find the values of x_n for which Newton's method <u>fails</u> due to there being oscillatory x-intercepts – i.e. $x_{n+2} = x_n$ where x_n is not already a solution of f. (2 marks)

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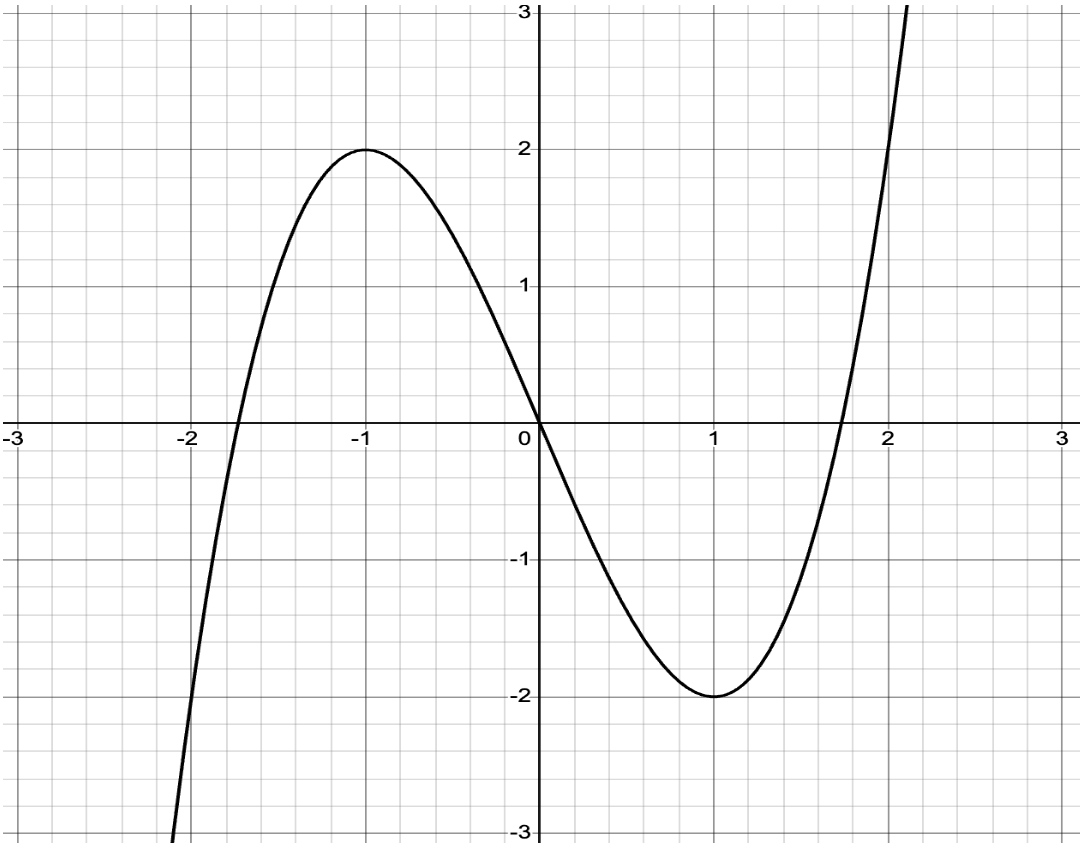
d. Sketch the tangent at each point $x = x_n$ for the values of x_n found in (c), labelling their x-intercepts. (1 mark)



Now consider the function, g, defined as:

$$g\colon R \to R,\ g(x) = x^{\frac{1}{3}}$$

e. Explain why that for any initial guess, $x_0 \neq 0$, Newton's method is unable to converge toward the x-intercept of g. (1 mark)

f. Now consider the initial guess $x_0 = 1$.

i. What is the value of x_4? (1 mark)

ii. Find the expression for the nth x-intercept, x_n, of Newton's method for g with $x_0 = 1$. (2 marks)

iii. Hence, or otherwise, find the value of x_{100} in the form a^b where $a, b \in Z^+$. (1 mark)